IBM Z DATATHON 2026 · THEME 1 REAL-TIME AI FOR CRITICAL DECISIONS + THEME 2 AI SECURED

SepsisWatch Z

Early sepsis warning for the ICU, running entirely on IBM LinuxONE (z17): predictive AI on the Telum II NNPA, generative AI with IBM Granite, and CPACF hardware security. Patient data never leaves the mainframe.

Team: Hemsy · Jhojan · Misael · Miguel

github.com/HemsyCA/DATAIBMZOS-IBM-Z-Datathon-2026 · Event code SSF2026

THE PROBLEM

Sepsis kills fast, and today's ICU alarms are too noisy to trust

~11M

sepsis-related deaths per year, about 1 in 5 deaths worldwide (Rudd et al., Lancet 2020)

every hour

of delayed treatment is associated with higher mortality (Kumar 2006; Seymour, NEJM 2017)

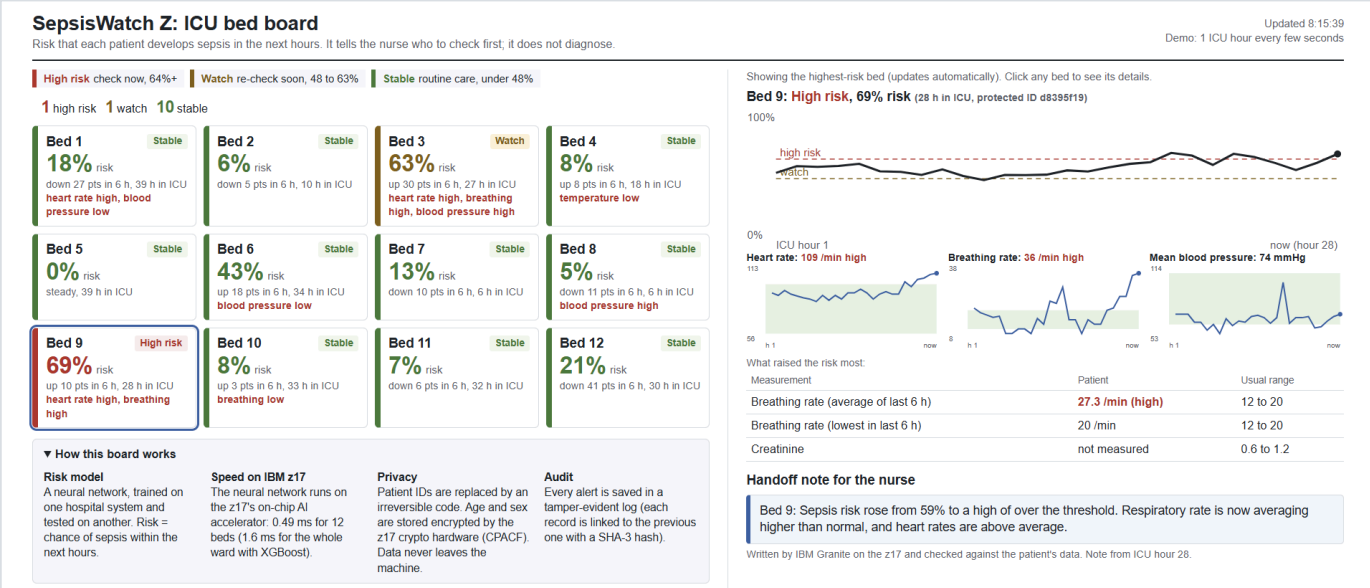
24%

of the time, SIRS ≥ 2 kept patients **without** sepsis in alarm on our test hospital → alarm fatigue

- Early signs are subtle and scattered across dozens of vitals and labs (lactate is missing in 97% of hours).
- ICUs with few specialists cannot watch every trend of every bed, every hour.
- Health data is highly sensitive: sending it to public-cloud AI is often not an option.

THE SOLUTION

One bed board: who to look at first, why, and a handoff note



Zoom: why the risk is high, and the IBM Granite handoff note (fact-checked)

What raised the risk most:

Measurement	Patient	Usual range
Breathing rate (average of last 6 h)	**27.3 /min (high)**	12 to 20
Breathing rate (lowest in last 6 h)	20 /min	12 to 20

Handoff note for the nurse

> Bed 9: Sepsis risk rose from 59% to a high of over the threshold. Respiratory rate is now averaging higher than normal, and heart rates are above average.

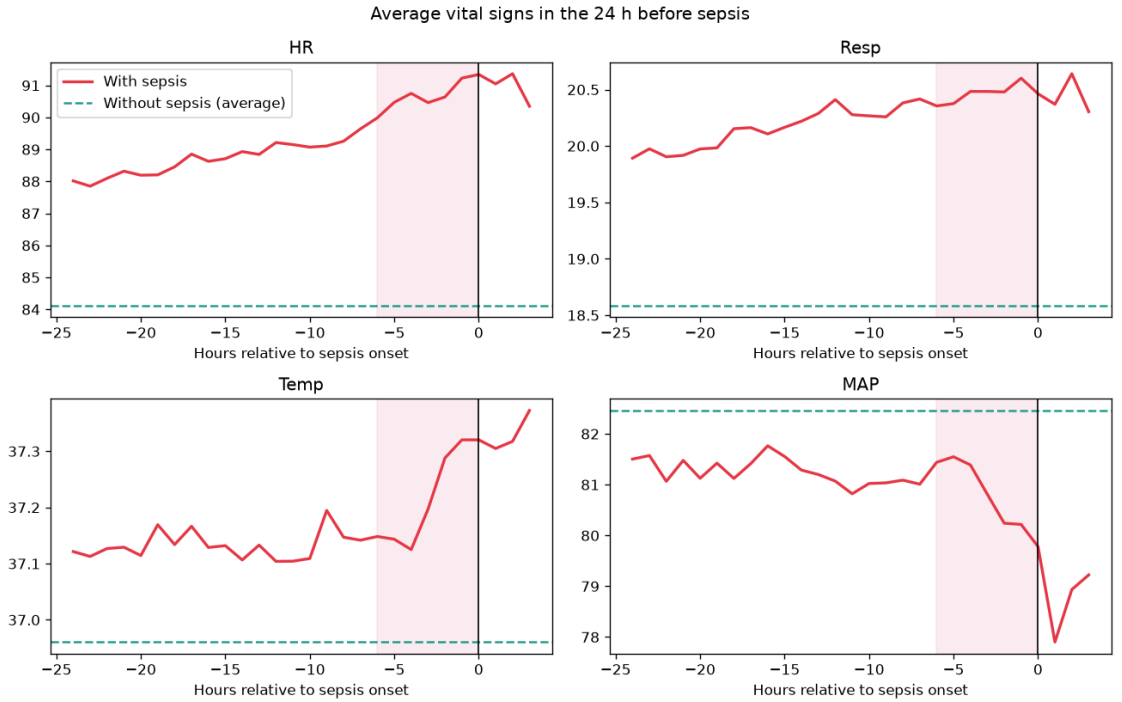
Written by IBM Granite on the z17 and checked against the patient's data. Note from ICU hour 28.

- **Every patient, every hour**: a risk score and a level: **Critical** / **Watch** / **Stable**.
- **Why this alert**: top clinical drivers (SHAP), physiology only.
- **Handoff note** by IBM Granite on the z17, **fact-checked** against the data before a nurse sees it.
- Whole ward scored in **~1.5 ms** (MLP on the NNPA + XGBoost).
- Decision support, not diagnosis: **clinicians always decide**.

DATA & METHOD

Real ICU data, validated on a hospital the model never saw

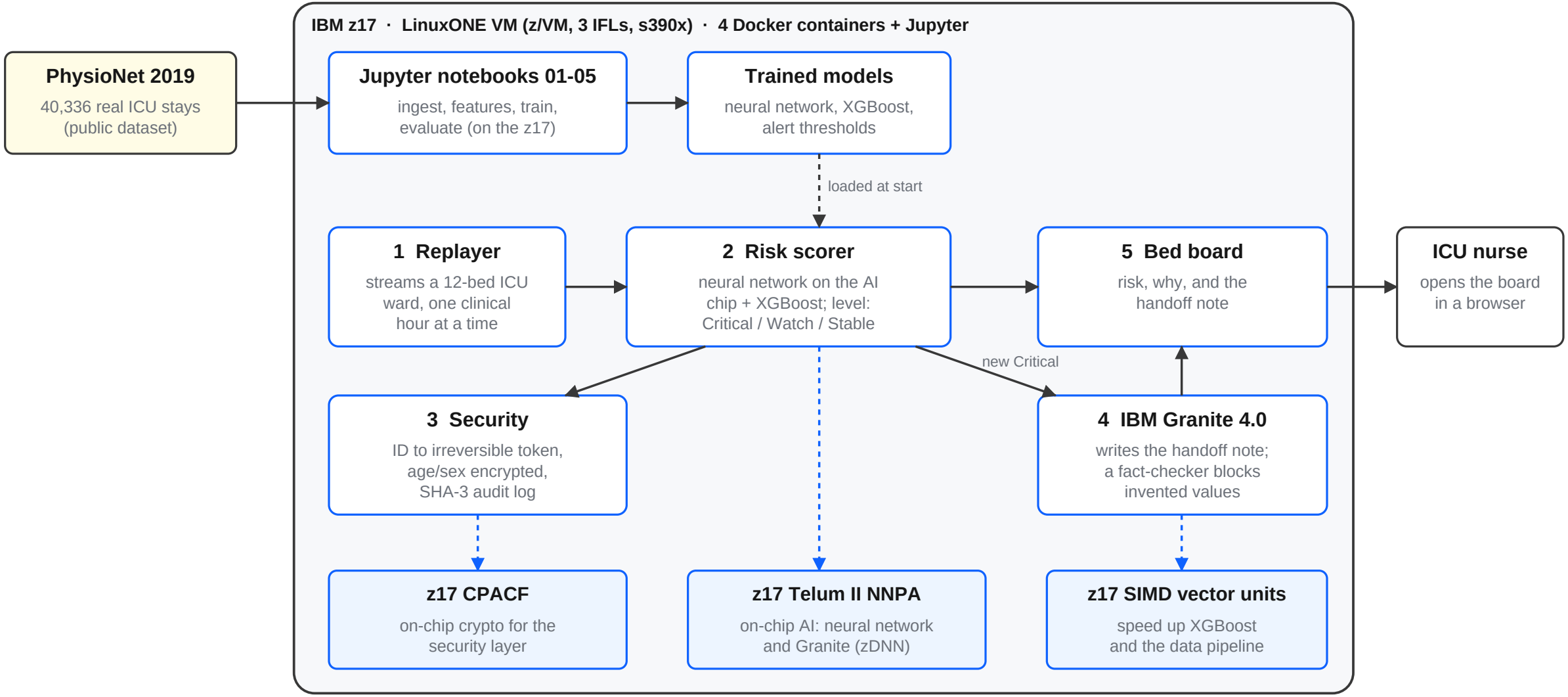
- **PhysioNet 2019 Sepsis Challenge**: 40,336 real ICU stays, 1,552,210 patient-hours, 40 variables.
- **Train on hospital A** (16,269 train / 4,067 validation patients) → **test on hospital B** (20,000 patients, 1,142 with sepsis).
- **140 hourly features** using only past data: last values, measurement patterns, 6-h trends, SIRS, qSOFA, shock index.
- **Models**: XGBoost + neural network (256-128-64) whose inference runs on the NNPA; SIRS/qSOFA baselines.
- **Alarm-aware thresholds**, chosen on validation: Critical ≤ 1 new false alarm per patient-day **and** ≤ 10% of non-sepsis hours.



Vital-sign trends before sepsis onset (notebook 01, run on the z17).

ARCHITECTURE

Everything runs inside one IBM z17: patient data never leaves



Detailed UML deployment and component diagram: appendix (slide 11) and docs/figures/architecture.png in the repository.

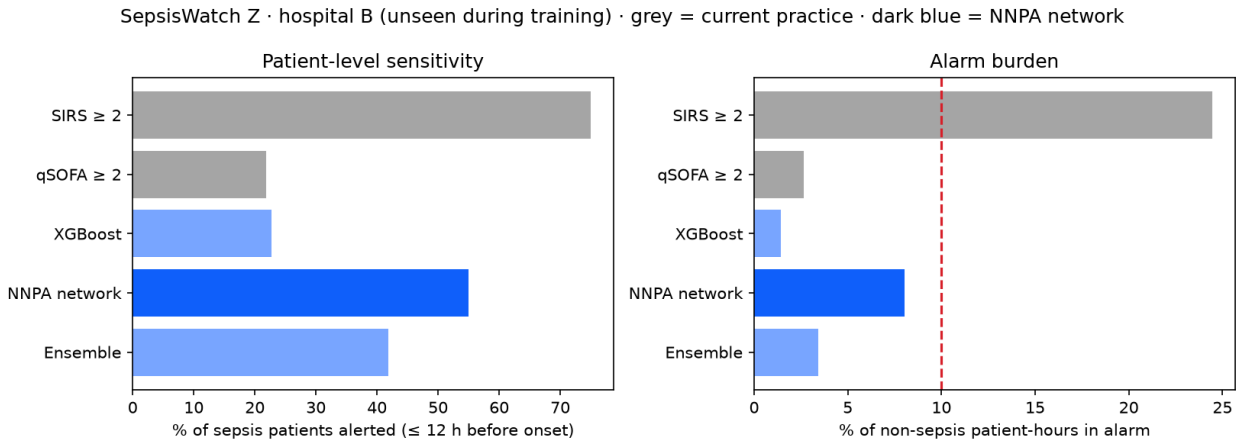
RESULTS · HOSPITAL B (UNSEEN)

Useful detection with one third of SIRS's alarm time

55%
of sepsis patients alerted up to 12 h before onset

8% vs 24%
non-sepsis patient-hours in alarm vs SIRS ≥ 2

−63%
new false alarms (0.49 vs 1.32 per patient-day)



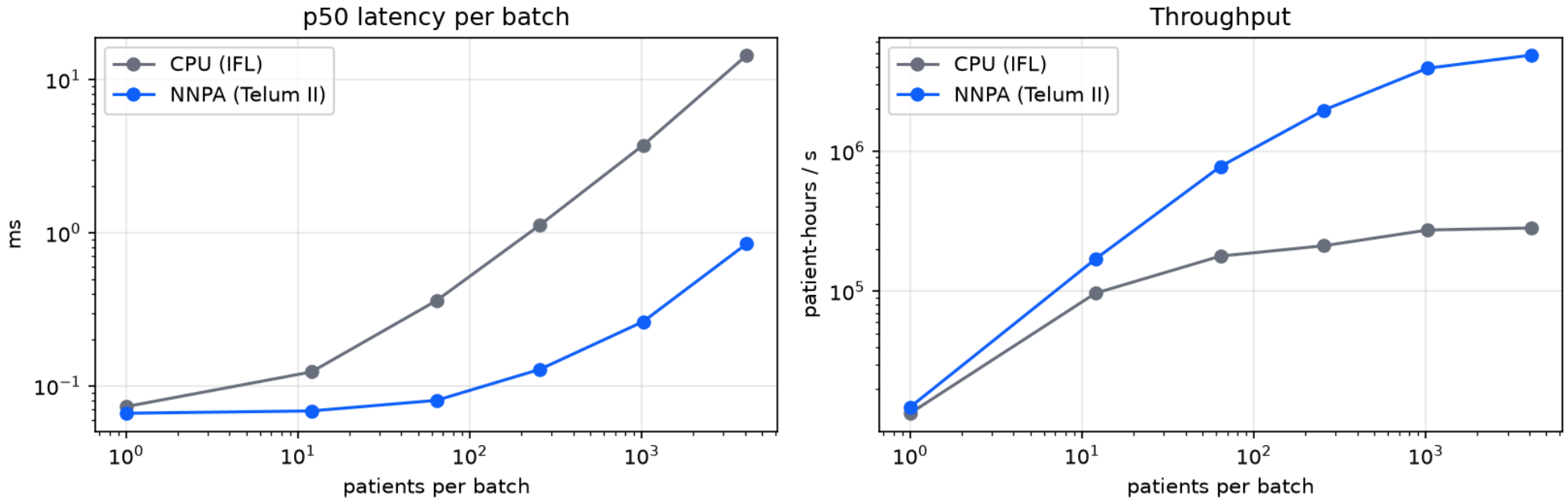
Model	Sens.	Alarm h	AUROC
SIRS ≥ 2	75.0%	24.4%	0.665
qSOFA ≥ 2	21.8%	2.7%	0.578
XGBoost	22.8%	1.4%	0.746
NNPA network	**55.0%**	**8.0%**	**0.711**
Ensemble	41.9%	3.4%	0.747

At near-SIRS sensitivity (68% vs 75%), the NNPA network still spends 40% less time in alarm (14.6% vs 24.4%). Model chosen on validation, not on the test set.

IBM Z PERFORMANCE

Telum II on-chip AI: faster at scale, identical decisions

SepsisWatch Z · MLP [256, 128, 64] (77,057 parameters) · hospital B · IBM z17



12×

whole hospital B (762k patient-hours): 0.22 s on NNPA vs 2.73 s on CPU

17×

4,096 patients per batch: 0.85 ms vs 14.4 ms · 12-bed ward in 0.07 ms

99.98%

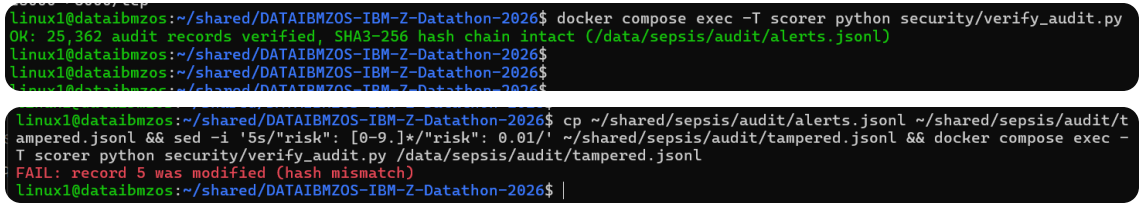
identical alert decisions CPU vs NNPA · same AUROC (0.711)

AI SECURED · TRUST BY DESIGN

Hardware security and a generative AI that cannot make things up

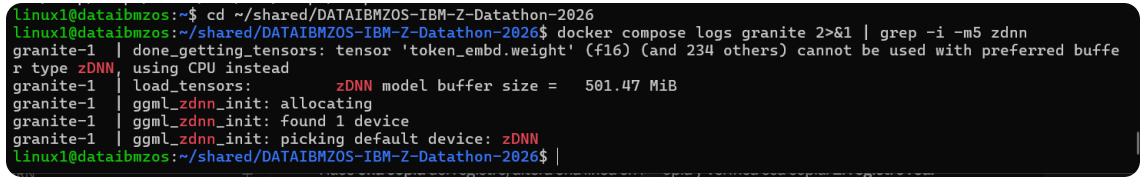
Security on IBM Z

- Patient IDs → **HMAC-SHA256 tokens**; key stays on the VM.
- Age/sex → **AES-256-GCM via OpenSSL → CPACF**.
- Every alert → **SHA3-256 hash chain**; edit one record and verification fails.
- No external API, no cloud LLM; ports bound to localhost.



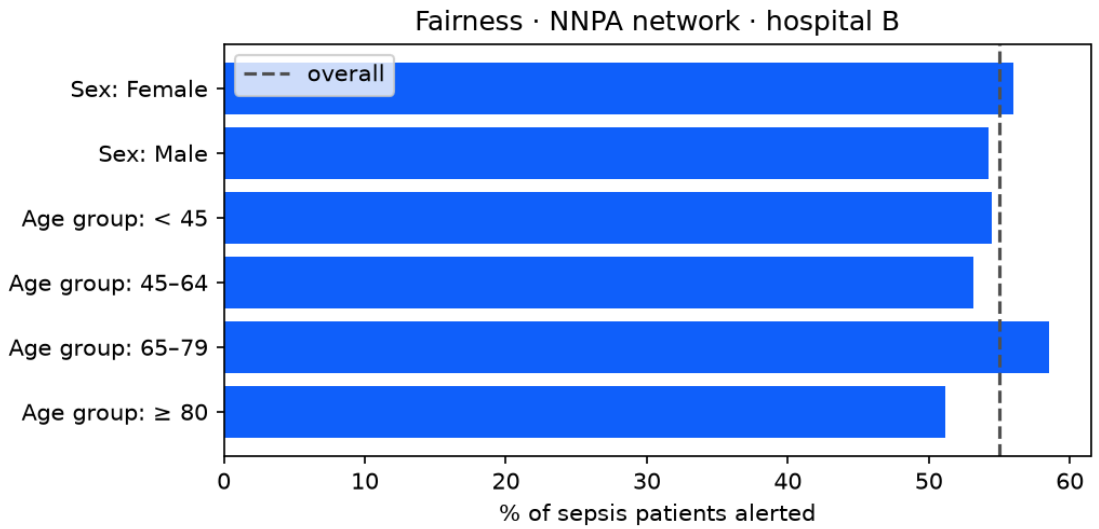
IBM Granite + fact-checker

- Granite 4.0 runs **on the z17** via llama.cpp's **zDNN** backend; about 7 s per note.
- Gets de-identified facts only; the bed number is set by the system.
- A note is rejected if it adds a number or topic not in the data, contradicts high/low, invents history or mentions drugs.
- Rejected → a deterministic template. **A nurse never sees an invented value.**



RESPONSIBLE AI

Fair across groups, honest about limits



Sensitivity by group, NNPA network, hospital B: gap 1.7 points by sex, 7.4 by age (lowest ≥ 80).

- Thresholds cap false alarms and alarm time to fight alarm fatigue.
- Explanations use physiology only; measurement-pattern features are never shown to nurses.
- A **risk score**, not a calibrated probability; decision support only.
- **Limits**: public 2019 US benchmark; replayed, not live; SIRS still catches more patients (75% vs 55%) at three times the alarm time; small Granite model.

IMPACT & NEXT STEPS

From a datathon prototype to a hospital pilot

Who benefits

- **ICU nurses & physicians**: know where to look first; alarms worth trusting.
- **Patients**: earlier attention to those deteriorating.
- **Hospitals**: on-premise AI with hardware encryption and an auditable trail; one IBM Z can serve many wards or hospitals.

Next steps

- Local validation and threshold tuning with partner hospitals.
- HL7 / FHIR integration with the EHR and bedside monitors.
- Larger or quantized Granite model for richer, faster notes.
- OpenShift on IBM Z for multi-hospital scale.

Built Oct 17–18, 2026 on IBM LinuxONE Community Cloud. Repo with 5 executed notebooks, 4 services and evidence screenshots: github.com/HemsyCA/DATAIBMZOS-IBM-Z-Datathon-2026

Telum II NNPA | zDNN | CPACF | z/VM | s390x containers | IBM Granite | PyTorch | XGBoost | FastAPI

APPENDIX · DETAILED ARCHITECTURE (UML DEPLOYMENT & COMPONENT DIAGRAM)

